

JAEHYUN LEE

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RESEARCH INTERESTS

Computer Graphics, Physics-Based Animation/Simulation, PDE, High-performance Computing, Numerical methods, Optimization

EDUCATION

University of Wisconsin-Madison

Wisconsin, USA

Ph.D. in Computer Sciences

Sep. 2024 – present

- Advisor: [Prof. Eftychios Sifakis](#)

Korea University

Seoul, Republic of Korea

M.S. in Computer Science and Engineering

Sep. 2021 – Feb. 2024

- Advisor: [Prof. JungHyun Han](#) and [Prof. Kiwon Um](#)
- GPA: 3.93/4.0

Korea University

Seoul, Republic of Korea

B.S. in Computer Science and Engineering (Double major)

Mar. 2019 – Feb. 2021

B.S. in Mechanical Engineering

Mar. 2015 – Feb. 2021

- Including 2 years of military service
- GPA: 3.98/4.0
- **Graduated with Great Honor (*Summa Cum Laude*)**

PUBLICATIONS

- Seung-wook Kim, HuiSeong Lee, **JaeHyun Lee**, Kiwon Um, JungHyun Han. “Dimension Expansion for Untangling Mass-Spring System.” In Computer Animation and Virtual Worlds (CAVW), Vol. 36, No. 6, November 2025, pp. e70079. [\[paper\]](#) [\[video\]](#)
- Heejo Jeong, Seung-wook Kim, **JaeHyun Lee**, Kiwon Um, Min Hyung Kee, JungHyun Han. “Momentum-preserving inversion alleviation for elastic material simulation.” In Computer Animation and Virtual Worlds (CAVW), Vol. 35, No. 3, May 2024, pp. e2249. [\[paper\]](#) [\[video\]](#)
- **JaeHyun Lee**, Seung-wook Kim, Kiwon Um, Min Hyung Kee, JungHyun Han. “Inversion alleviation for stable elastic body simulation.” In Computer Animation and Virtual Worlds (CAVW), Vol. 34, No. 3-4, May 2023, pp. e2183. [\[paper\]](#) [\[video\]](#)

FEATURED PROJECTS

Convolutional Solver

University of Wisconsin–Madison

Researcher

Oct. 2024 – Present

- Developed a general-purpose 2nd-order elliptic PDE solver in PyTorch with CUDA acceleration, supporting irregular geometries and mixed boundary conditions. The solver has been applied to computer graphics problems including fluid, thermal, and inverse simulations.
- Project is actively in progress; [intermediate results](#) available.

General Purpose Material Point Method (MPM) Engine

Korea University

Researcher

Oct. 2023 – present

- Developed C++, CUDA-based state-of-the-art MPM framework, with visualization system using OpenGL. [\[code\]](#)

LG Electronics: Air Conditioning Airflow Simulation Visualization System

Korea University

Project Assistant

Mar. 2022 – Aug. 2022

- Contributed to the project by implementing Python-based, GPU-accelerated real-time airflow simulator visualized with volume rendering. The project won the **first prize** among 489 teams. [\[code\]](#) [\[video\]](#)

Collision Detection for Constrained Projective Dynamics (CPD)

Korea University

Researcher

Dec. 2020 – May. 2021

- Implemented tetrahedral collision detection module for ACM Transactions on Graphics 2021 paper titled ‘Constrained Projective Dynamics: Real-Time Simulation of Deformable Objects with Energy-Momentum Conservation’. [\[paper\]](#) [\[video\]](#) [\[code\]](#)

TEACHING

Parallel & Throughput- Optimized Programming (CS557) <i>Teaching Assistant</i>	University of Wisconsin-Madison <i>Spring 2026</i>
Computer Graphics (CS559) <i>Teaching Assistant</i>	University of Wisconsin-Madison <i>Spring 2025, Fall 2025</i>
Programming III (CS400) <i>Teaching Assistant</i>	University of Wisconsin-Madison <i>Fall 2024</i>
Computer Graphics (COSE331) <i>Teaching Assistant</i>	Korea University <i>Spring 2022</i>

SCHOLARSHIPS

Kwanjeong Educational Foundation Scholarship, Kwanjeong Educational Foundation	<i>Spring 2022 – Fall 2023</i>
Teaching Assistant Scholarship, Korea University	<i>Spring 2022</i>
Research Scholarships, Korea University	<i>Fall 2021, Fall 2022</i>
National Science and Engineering Scholarship, Ministry of Science and ICT	<i>Spring 2019 – Fall 2020</i>
Special Scholarships, Korea University	<i>Spring, Fall 2018</i>

HONORS AND AWARDS

Best Industry-Academic Project Award, Ministry of Trade, Industry and Energy	<i>Nov 2023</i>
Best Research award, Korea Electronics Association	<i>Feb 2022, Dec 2022, Aug 2023</i>
Great Honor, Korea University	<i>Graduation</i>
President's List, Korea University	<i>Fall 2018 – Spring 2019</i>
Dean's List, Korea University	<i>Spring 2018</i>
Semester High Honors, Korea University	<i>Spring 2017 – Spring 2020</i>

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript

APIs: OpenGL, CUDA, Three.JS ,OpenMP

Other Tools and Libraries: Git, Eigen, Partio, ImGui, Assimp, PyTorch, Fusion360, CMake, Taichi Lang, Blender

LANGUAGE LEVEL

Korean: Native

English: Fluent